

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026

Biology — Class IX | Paper: SSC-I

**Syllabus: Ch 1 — The Science of Biology | Ch 2 — Biodiversity | Ch 3 — The Cell | Ch 4 — Cell Cycle
| Ch 5 — Tissues, Organs & Organ Systems | Ch 6 — Molecular Biology | Ch 7 — Metabolism (Sec
7.1-7.6)**

ANSWER KEY (Version 4)

Total Marks: 65

Time: 2h 30m

Version: 4

SECTION A — Multiple Choice Questions Answer Key (12 × 1 = 12 Marks)

Q1. Mosquito with zebra-like stripes that spreads Dengue fever: ✓ **C — Aedes aegypti** **1 mark**

The textbook states: “Dengue Fever... is caused by a Dengue virus and is transmitted by mosquito **Aedes aegypti**, which has zebra like white and black stripes on its body.” This is different from **Anopheles**, which transmits malaria. Prevention relies on personal protection from mosquito bites and controlling mosquito breeding sites.

Q2. Scientist who studies effects of low temperature on living organisms to preserve cells/tissues: ✓ **D — Cryobiologist** **1 mark**

The textbook's sidebox on emerging careers lists: “**Cryobiologists** (study of effects of low temperature on living organisms).” Bioinformaticians apply computer skills to life-science problems; biomedical engineers develop devices/equipment for health; astrobiologists study the effects of outer space on living organisms — all different specialisations from the same sidebox.

Q3. Biologist (1656-1708) who introduced the ranks of class and species: ✓ **C — Tournefort** **1 mark**

The textbook's History of Classification states: “**Tournefort (1656-1708)**: He introduced the taxa of class and species.” Andrea Caesalpino (1519-1603) divided plants into fifteen genera; John Ray (1627-1705) published works on classifying plants; Linnaeus (1707-1778) later grouped species by physical characteristics and introduced binomial nomenclature.

Q4. What Robert Hooke observed in cork tissue in 1665: ✓ **C — Empty chambers with thick outer coverings** **1 mark**

The textbook states: “In 1665, Robert Hooke discovered cell when he examined a thin slice of cork tissue under a compound microscope. He observed cells as **empty chambers with thick outer coverings**.” The nucleus was only discovered later (1831) as microscope lens quality improved.

Q5. Fluid part of the chloroplast surrounding the grana: ✓ **B — Stroma** **1 mark**

The textbook states: “Many thylakoids stack to form granum... The fluid part of chloroplast is called **stroma**.” The matrix is the fluid inside a mitochondrion (not a chloroplast); cristae are the folds of a mitochondrion's inner membrane; lumen refers to the internal space of structures like the Golgi apparatus.

Q6. What happens mainly during the G2 phase of interphase: ✓ **B — Forms spindle proteins, stores energy** **1 mark**

The textbook states: “**G2 phase**: During this phase cell prepares for mitotic phase. Cell forms proteins especially those required for the formation of spindle fibres. Also cell accumulates energy to complete the division process. More mitochondria are formed.” DNA is replicated in the S phase (not G2), and the cell physically divides only in the M phase.

Q7. Second event of meiosis (besides crossing over) that produces genetic variation: ✓ **B — Independent combination of chromosomes** 1 mark

The textbook states: “Crossing over and **independently formed combinations of chromosomes** in gametes are two important events of meiosis. Both crossing over and chromosomal combination provide basis of variations.” DNA replication and synapsis occur in every meiotic division and do not themselves create new gene combinations the way independent assortment does.

Q8. Primary functions of xylem and phloem: ✓ **B — Xylem—support & transport of water; Phloem—transport of sugars** 1 mark

This is confirmed directly by the textbook's own exercise: xylem provides **support and transport of water**, while phloem provides **transport of sugars** (prepared food) from the leaves to the rest of the plant.

Q9. Why respiratory and circulatory systems work closely together: ✓ **B — Circulatory system delivers O₂ from lungs to cells, returns CO₂** 1 mark

The textbook states: “The respiratory system and the circulatory system work closely together to deliver oxygen to cells and get rid of the carbon dioxide that these cells produce. The circulatory system picks up oxygen in the lungs and drops it off in the tissues, then performs the reverse function for carbon dioxide.”

Q10. Test confirming starch by a dark blue-black colour: ✓ **C — Iodine test** 1 mark

Per STEAM Activity 6.3: “**Iodine test for Starch**... Dark blue black coloration is produced... Starch is confirmed.” The Biuret test detects protein (pink/violet); Benedict's test detects reducing sugars/glucose (colour change on heating); the emulsion test detects lipids (cloudy white suspension).

Q11. Percentage of cytosine if adenine is 30% in DNA: ✓ **A — 20%** 1 mark

The Watson & Crick model states: “The amount of adenine and thymine is always equal and amount of cytosine and guanine is always equal.” If adenine (A) = 30%, then thymine (T) = 30% too, so A + T = 60%. The remaining 40% is shared equally between cytosine (C) and guanine (G), so **C = G = 20%**.

Q12. Permanent damage to an enzyme's shape/active site from excess heat: ✓ **C — Denaturation** 1 mark

The textbook states: “At very high temperature atoms of the enzyme molecule start vibrating vigorously so, globular structure and active site of enzyme is damaged and it is called **denatured**.” This differs from saturation (all active sites occupied by substrate) and inhibition (a molecule blocking the enzyme without heat damage).

SECTION B — Short Questions Answer Key (11 × 3 = 33 Marks)

2(i) — Characteristics of a good hypothesis 3 marks

A scientific hypothesis should have the following four characteristics: 0.75 each

- (a) It should be a **general statement**.
- (b) It should be a **tentative idea** (not fixed/final).
- (c) It should **agree with the available observations**.
- (d) It should be **testable and potentially falsifiable**.

OR

Dengue fever: Caused by the **Dengue virus**, transmitted by the mosquito **Aedes aegypti**, which has zebra-like black and white stripes on its body. 1 mark

Typical Dengue haemorrhagic fever is characterized by high-grade fever, bleeding from the nose, blood in urine, and an enlarged liver; there is no specific antiviral drug available. 1 mark

Prevention: Personal protection from mosquito bites, along with measures to prevent mosquito breeding. 1 mark

2(ii) — Early classifiers before Linnaeus 3 marks

Andrea Caesalpino (1519-1603): Divided plants into fifteen groups and called them genera. 1 mark

John Ray (1627-1705): Published important works on the classification of plants. 1 mark

Tournefort (1656-1708): Introduced the taxa (ranks) of class and species. 1 mark

OR

Kingdom Protista includes eukaryotic organisms with unicellular or colonial organization and is a very diverse group. It includes three types: 1 mark each

(a) **Animal-like protists** called **protozoa**, e.g. *Amoeba*.

(b) **Plant-like protists** called **algae**, e.g. *Euglena*.

(c) **Fungus-like protists**, e.g. **slime molds**.

2(iii) — Discovery of the cell and the nucleus 3 marks

In 1665, **Robert Hooke** discovered the cell when he examined a thin slice of cork tissue under a compound microscope. 1 mark

He observed cells as **empty chambers with thick outer coverings**. 1 mark

The quality of microscope lenses improved greatly in the nineteenth century, which led to the discovery of the **cell nucleus in 1831**, and many cytoplasmic organelles in the years that followed. 1 mark

OR

Chloroplast envelope: Two membranes surrounding the chloroplast, forming the chloroplast envelope; chloroplasts have their own DNA and ribosomes and can multiply within the cell. 1 mark

Grana: Many hollow, coin-like membranous structures (thylakoids) stack together to form granum (plural: grana); these membranes contain chlorophyll and other photosynthetic pigments. 1 mark

Stroma: The fluid part of the chloroplast surrounding the grana. 1 mark

2(iv) — Chemical composition of the cell membrane 3 marks

Chemically, the cell membrane is composed of **proteins (60-80%)**, **phospholipids (20-40%)**, and traces of carbohydrates. 1.5 marks

According to the fluid mosaic model, the membrane consists of a double layer (bilayer) of phospholipids in which protein molecules float like icebergs in a fluid sea of phospholipids. 0.5 mark

Cholesterol: Cell membranes of eukaryotes also contain cholesterol, which prevents stiffening of the cell membrane and is required for the fusion of secretory vesicles with the membrane. 1 mark

OR

Chromoplasts: Coloured other than green (red, pink, yellow, blue, purple, etc.); found in flower petals to attract pollinating insects, and in the walls of ripened fruits to attract birds and animals that help in seed dispersal. 1.5 marks

Leucoplasts: Non-pigmented plastids; food-storing organelles usually found in roots, bulbs, and stem tubers; store carbohydrates, proteins, or lipids. 1.5 marks

2(v) — Crossing over and independent combination causing variation 3 marks

Variations are the differences among members of the same species, and are necessary for survival in an always-changing environment. 0.5 mark

Crossing over: During prophase I of meiosis, non-sister chromatids of homologous chromosomes exchange segments of genetic material at points called chiasmata — this creates new gene combinations on each chromatid. 1.25 marks

Independent combination: Chromosomes are also independently (randomly) combined into gametes during meiosis, producing new combinations of whole chromosomes. 1.25 marks

Together, crossing over and independent chromosome combination provide the basis for genetic variation in gametes and, ultimately, in the zygote.

OR

A cell with 24 chromosomes (diploid, somatic) replicates its DNA during S phase, so by **late prophase of mitosis** the cell still has **24 chromosomes**, but each chromosome now consists of **2 sister chromatids** (joined at one centromere). 1.5 marks

Chromosome *number* does not change during S phase — only the number of chromatids per chromosome doubles. After mitosis is complete, each daughter cell receives **24 chromosomes** (each again with a single chromatid), the same number as the original parent cell. 1.5 marks

2(vi) — Phragmoplast and cell wall formation in plant cytokinesis 3 marks

In a plant cell, spindle fibres in the equator region form a structure called the **phragmoplast**. 1 mark

The **Golgi apparatus** forms vesicles which first appear in the centre of the phragmoplast and then grow outward toward the equatorial plane. 1 mark

The contents of these vesicles form the **middle lamella and primary walls** of the two daughter cells; some cells later also form secondary walls. 1 mark

OR

During the G2 phase of interphase: 0.75 mark each

- (a) The cell prepares for the mitotic (M) phase.
- (b) It forms proteins, especially those required for the formation of spindle fibres.
- (c) The cell accumulates energy needed to complete the division process.
- (d) More mitochondria are formed.

2(vii) — Digestive system organs and homeostatic role 3 marks

Organs: Mouth, salivary glands, oesophagus, stomach, liver, gallbladder, exocrine pancreas, small intestine, and large intestine. 1.5 marks

Function: Processes foods and absorbs nutrients, minerals, vitamins, and water.

Role in homeostasis: Maintains the levels of nutrients in the blood. 1.5 marks

OR

“Organ systems working together” means that different organ systems cooperate to accomplish a shared task, keeping the whole body running smoothly. 1 mark

Example: The respiratory system and the circulatory system work closely together to deliver oxygen to cells and remove carbon dioxide. The circulatory system picks up oxygen in the lungs and drops it off in the tissues, then performs the reverse function to collect CO₂ and carry it back to the lungs for removal.

2 marks

2(viii) — Six major bioelements and their percentages 3 marks

The six major bioelements make up 99% of protoplasm: 0.5 mark each

Oxygen — 65% | Carbon — 18% | Hydrogen — 10% | Nitrogen — 3% | Calcium — 2% | Phosphorus — 1%

OR

Purines: Nitrogen bases with a **double-ring** structure — **Adenine** and **Guanine**. 1.5 marks

Pyrimidines: Nitrogen bases with a **single-ring** structure — **Cytosine**, **Thymine**, and **Uracil**. 1.5 marks

2(ix) — Benefits of cellulose fibre 3 marks

Any THREE of the following, 1 mark each :

- (a) Acts as fibre/roughage in the human diet, helping food move efficiently through the digestive tract and preventing constipation.
- (b) Absorbs away toxic chemicals that might be present in food.
- (c) Used in the paper industry.
- (d) Helps lower blood cholesterol and regulate blood pressure.

OR

Level	Description
Primary structure	A linear polypeptide with a specific number, type, and sequence of amino acids.
Secondary structure	Formed upon folding of the polypeptide with extra (hydrogen) bonds.
Tertiary structure	Further folding of the secondary structure with even more extra bonds, often into a globular form.
Quaternary structure	Two or more polypeptides, each with tertiary folding, bonded together.

0.75 mark each

2(x) — Lock-and-key vs induced-fit models 3 marks

Lock-and-key model: The enzyme's active site is considered to have a fixed, **rigid** shape that matches the substrate exactly, just as a key fits only one specific lock. 1.5 marks

Induced-fit model: The active site is **flexible** — it slightly changes/molds its shape around the substrate as the substrate binds, giving a better fit once binding begins. 1.5 marks

OR

ATP (Adenosine Triphosphate) is called the “**energy currency**” of the cell because it is an energy-rich compound that can store and release energy, and can be used much like money to “pay for” cellular work.

1 mark

ATP-ADP cycle: ATP can be converted to ADP and inorganic phosphate by hydrolysis, releasing energy for cellular work; the third phosphate group separates and remains in the cell in inorganic form. 1 mark

ADP and phosphate can be converted back into ATP by condensation, using energy released from catabolism (or light energy absorbed by plants). 1 mark

2(xi) — Energy of activation and the role of enzymes 3 marks

Energy of activation: The minimum amount of extra energy required to activate a molecule so that it can react, or to start a reaction. 1.5 marks

Enzymes **lower the activation energy** needed for a reaction, so that biochemical reactions can take place at normal body temperature. They do this by changing the shape and charge of the substrate, and may provide an entirely different reaction pathway/mechanism to lower the activation energy required.

1.5 marks

OR

In the presence of enough enzyme molecules, the rate of an enzyme-controlled reaction is **directly proportional** to substrate concentration, up to a certain limit. 1.5 marks

Saturation of active sites: Beyond that limit, a point is reached where a further increase in substrate concentration has no further effect on the rate of reaction. This happens because extra substrate molecules in the medium have already occupied **all** the active sites of the available enzyme molecules. 1.5 marks

SECTION C — Long Questions Answer Key (4 × 5 = 20 Marks)

3(i) — Science as a collaborative field + emerging biology careers 5 marks

(a) Science is a collaborative field: Scientists from around the world team up to share ideas and make progress in research. When researchers from different fields work together to create new scientific knowledge, it is called **interdisciplinary research collaboration**. This is important because they can work on research, find solutions, and use what they learn to solve problems and discover new things. 1 mark

Two examples from the textbook (any two): 2 marks

(i) **Cognitive Science** combines knowledge from neurology, psychology, anthropology, linguistics, environmental science, engineering, pharmacology, and statistics.

(ii) **Women's Studies** combines what we know about gender, history, literature, and biology.

(iii) **Public health** combines information from medicine, sociology, and psychology. (The International Space Station is also given as an example of scientists from different countries teaming up.) **(b) Any THREE emerging careers** (any three), 2 marks:

- **Bioinformaticians** — apply computer skills to solve problems in the life sciences.
- **Biomedical engineers** — develop new devices and equipment for improving human health.
- **Astrobiologists** — study the effects of outer space on living organisms.
- **Cryobiologists** — study the effects of low temperature on living organisms.

OR

(a) FOUR living characteristics of viruses (any four), 4 marks (1 each):

- They occur in different varieties.
- They have their own genetic material in the form of either RNA or DNA.
- They reproduce using the material of the host cell they infect.
- They enter the cells of living organisms and cause diseases.

THREE non-living characteristics (any three from the remaining marks):

- They lack cellular structure and an enzyme system.
- They can be crystallized and stored in a bottle.
- They do not respire.
- They behave as non-living, inert infectious particles outside the host. **(b)** Viruses are at the borderline of living and non-living things. So, they are not included in any domain or kingdom under modern classification. 1 mark

3(ii) — Mitochondria structure + cilia and flagella 5 marks

(a) Mitochondria: Found in all aerobic eukaryotic cells; they are **double membrane-bound** structures.

1 mark

The outer membrane is smooth, while the **inner membrane folds inward to form finger-like projections called cristae**, which increase the surface area available for respiration. The fluid inside the mitochondrion is called the **matrix**. 1 mark

Significance of own DNA/ribosomes: Mitochondria have their own DNA and ribosomes, which allows them to multiply within the cell independently. They produce energy in the form of ATP, which is why they are called the “power house of the cell.” 1 mark **(b) Cilia and flagella:** Hair-like (cilia) or whip-like (flagella) extensions found on some eukaryotic cells. Both consist of **nine pairs of microtubules surrounding a single central pair of microtubules**, and are anchored to the cell by a **basal body**. Their function is **movement**. 2 marks

OR

(a) Formation and function of a lysosome: 3 marks

The digestive enzymes contained in lysosomes are first synthesized on the **RER**, then transported to the **Golgi apparatus** for processing. Lysosomes then **bud off from the Golgi apparatus** with their processed enzymes.

During intracellular digestion, the cell takes up material from outside as a **food vacuole**. When a lysosome **fuses** with this food vacuole, the lysosomal enzymes act on the complex food substances and convert them into simple forms that the cell can use. Lysosomes also engulf and digest the cell's own unwanted organelles — a process called autophagy. **(b) Role of the endoplasmic reticulum:** The ER forms a complex, interconnected network of membranes throughout the cytoplasm, which provides **mechanical support** to the cell and is also involved in the **transport of substances within the cell**. 2 marks

3(iii) — Fruit fly meiosis + bivalent/tetrad 5 marks

(a) The fruit fly has a diploid chromosome number of 8. Since meiosis is a **reduction division** that halves the chromosome number, each gamete produced would contain **4 chromosomes** (half of 8). 1.5 marks

This reduction is necessary because, during fertilization, a haploid sperm (4 chromosomes) fuses with a haploid egg (4 chromosomes) to restore the diploid number (8) in the zygote. If gametes were not haploid, the chromosome number would double with every generation; meiosis ensures that the chromosome number specific to each species stays constant across generations. 1.5 marks **(b) Bivalent (tetrad):**

During prophase I of meiosis, homologous chromosomes line up point to point and pair with each other — this pairing (synapsis) forms a **bivalent**, also called a tetrad. 1 mark

Since each chromosome has already replicated (S phase) into 2 sister chromatids before meiosis begins, a bivalent contains **4 chromatids** (hence “tetrad”) and **2 centromeres** (one per chromosome). 1 mark

OR

Meiosis II compared to mitosis — any THREE similarities, ~1.67 marks each:

(i) Meiosis II is divided into the **same four named phases** as mitosis: prophase, metaphase, anaphase, and telophase.

(ii) In both processes, **sister chromatids separate** at anaphase — the centromere divides and spindle fibres pull the individual chromatids to opposite poles.

(iii) In both, chromosomes line up **individually** at the equator during metaphase (not in homologous pairs as in meiosis I), each attached to spindle fibres from both poles; also, no further DNA replication occurs before either division begins (interphase II has no S phase, just like the short gap before mitosis in some cycles).

3(iv) — DNA vs RNA + Chargaff's rule calculation 5 marks**(a) Any FIVE differences between DNA and RNA:** 3 marks (0.6 each)

DNA	RNA
Double stranded (helix)	Single stranded
Pentose sugar is deoxyribose	Pentose sugar is ribose
Contains thymine as a base	Contains uracil instead of thymine
Found mainly in the nucleus (as chromosomes)	Found in the nucleolus and cytoplasm (mRNA, tRNA, rRNA)
Stores and carries hereditary information (genes)	Acts as an intermediate; tool for translation/protein synthesis
Bases pair with a complementary strand (hydrogen bonds)	Nitrogen bases are overhanging (not paired to a complementary strand)

(b) Chargaff's rule calculation: Given Guanine (G) = 20%. 2 marksSince the amount of cytosine always equals the amount of guanine: **Cytosine (C) = 20%**.So $G + C = 40\%$, leaving $100\% - 40\% = 60\%$ for A + T combined.Since adenine always equals thymine: **Adenine (A) = Thymine (T) = 30% each.****OR****(a) Intracellular vs extracellular enzymes:** 2 marks**Intracellular enzymes** stay inside the cell to speed up reactions there, e.g. the enzymes of glycolysis working in the cytoplasm.**Extracellular enzymes** are made inside the cell but released outside it to do their work, e.g. fungi and bacteria release extracellular enzymes to digest their food outside their own body. **(b) Regulation of****enzyme production and activity:** 3 marksThe **synthesis** of enzymes within a cell can be increased or decreased by the cell according to its requirements. 1.5 marksEnzyme **activity** (once made) can also be regulated — by inhibitors or activators, coming from either within or from outside the cell. 1.5 marks